Charge and the electric car

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Explain charge interactions and trace an electric car's energy transfers.

SUCCESS CRITERIA - I can...

1. Predict attraction or repulsion from charge signs.
2. Distinguish charge from energy.
3. Trace energy from a battery to moving wheels.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

01 Warm-up: bring back an earlier idea

An ideal blackbody's kelvin temperature becomes 3 times as large. How does its peak wavelength change?

Marzano check - circle: ① ② ③

02 Worked example: explain the model

Reference: $E_{electrical} = E_{mechanical} + E_{thermal}$ + other transfers. Energy accounting for a motor with no net energy storage. A battery transfers energy; charge already exists in the wires.

Worked reasoning: Rubbing can transfer electrons between materials. If one isolated object gains charge -3 units and another loses the same electrons, their charge changes add to zero.

Your turn: Draw before-and-after charge accounts for both objects. Distinguish transfer from creation of charge.



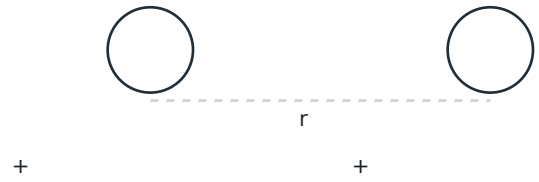
Label charges and separation.

Object A Object B

Marzano check - circle: ① ② ③

03 Read and interpret the model

Draw force arrows for two positive charges, then for a positive and a negative charge.

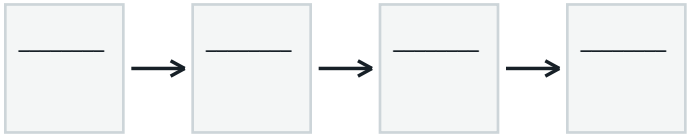


Each circle represents one of the labeled objects or charges; r is their center-to-center separation.

Marzano check - circle: ① ② ③

04 Apply the physics

A battery-powered model car begins moving. Put these stages in a sensible energy chain: car motion, stored chemical energy, electrical transfer and motor rotation. Use arrows and add one branch for energy transferred elsewhere.



Marzano check - circle: ① ② ③

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05 Explain your reasoning

A student says the motor uses up the current so less returns to the battery. Explain the mistake for a steady series circuit.

Marzano check - circle: 1 2 3

06 Review: connect an earlier idea

Two wave displacements at one point are -4 mm and +8 mm. Find the resultant and state whether they reinforce or partly cancel.
Show the signed contributions and their sum.

A

B

Sum

Add signed displacements; label units.

Marzano check - circle: 1 2 3

07 Exit check

Predict the interaction between two negative charges.



Label charges and separation.

Object A Object B

Marzano check - circle: 1 2 3

08 STRETCH: test the model

Draw one attraction between opposite charges and one repulsion between like charges. Explain whether attraction alone proves both objects carry opposite net charges.



Label charges and separation.

Object A Object B

Marzano check - circle: 1 2 3